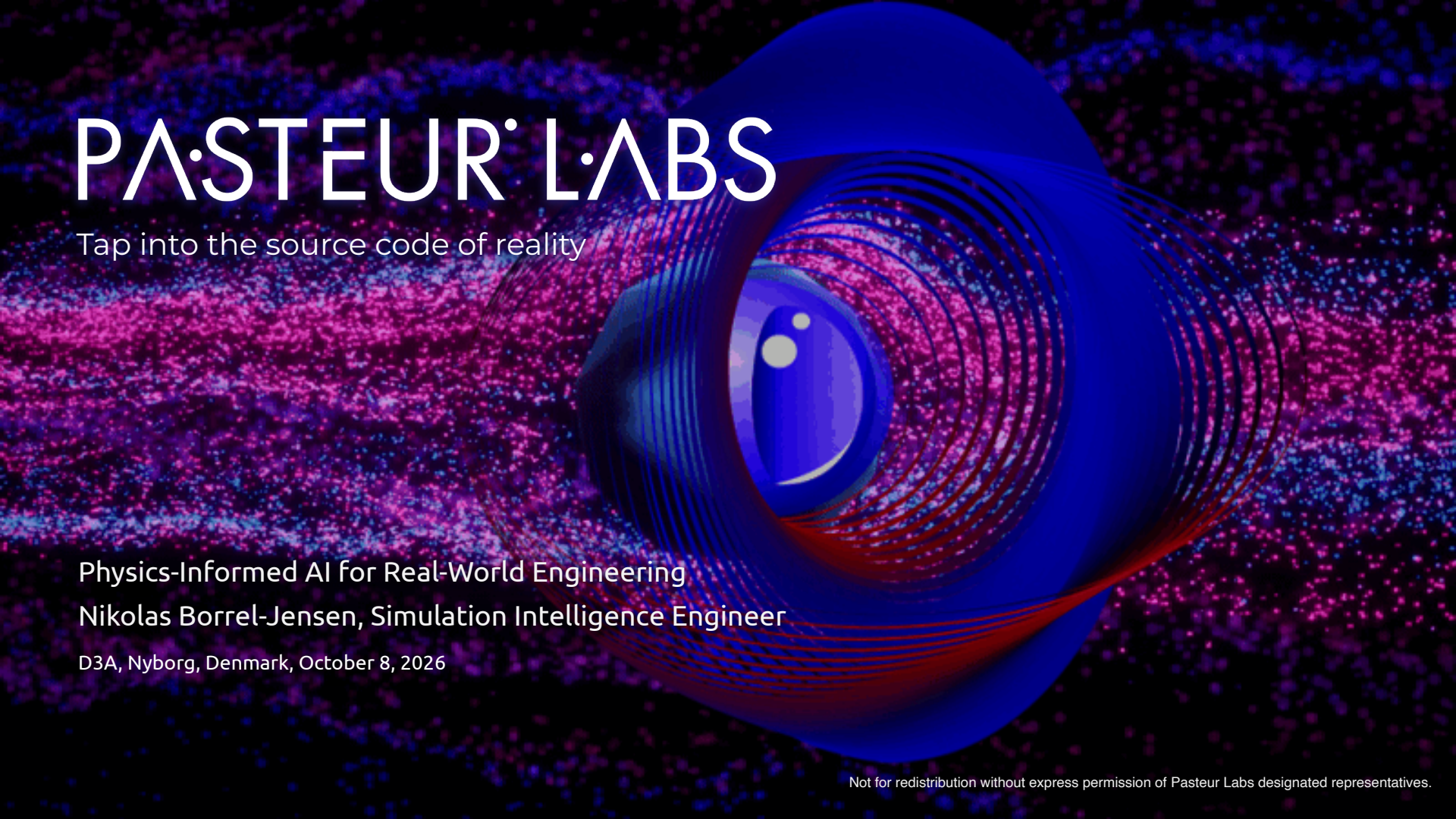


PASTEUR LABS

The background features a large, stylized blue eye-like shape with concentric rings, set against a dark field with vibrant pink and blue particle trails that create a sense of motion and depth.

Tap into the source code of reality

Physics-Informed AI for Real-World Engineering

Nikolas Borrel-Jensen, Simulation Intelligence Engineer

D3A, Nyborg, Denmark, October 8, 2026

Program – D3A, Nyborg, Denmark, October 8, 2026

Thursday 8 October 11.00

11:10 - 11:20 **Landscape framing** (Nikolas Borrel-Jensen)

11:20 - 11:40 **2 lightning talk** (Niels Skovgaard Jensen and Jakob Tanderup)

11:40 - 12:10 **Live tutorial** (Dion Haefner)

12:10 - 12:30 **Structured discussion** (Allan P. Engsig-Karup)



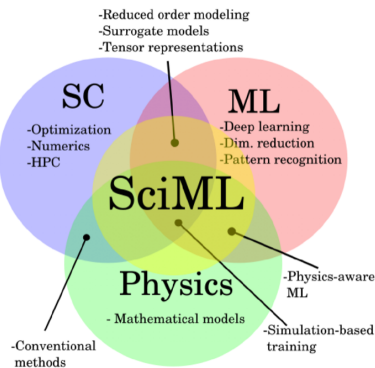
GitHub repo

Organizers



Physics-Informed AI for Real-World Engineering: a slice through the landscape in 10 minutes

Scientific Machine Learning (SciML) (2018 –)



Scientific Machine Learning (SciML) (coined at a workshop of the US Department of Energy in January 2018., cf. USDOES report, 2019, <https://www.osti.gov/biblio/1478744/>) : SciML is an emerging discipline within the data science community. SciML seeks to address

- domain-specific data challenges and extract insights from scientific data sets.
- SciML draws on tools from both machine learning and scientific computing to **develop new methods for scalable, domain-aware, robust, reliable, and interpretable learning and data analysis**,

Integrating Physics-Based Modeling With Machine Learning: A Survey

JARED WILLARD* and XIAOWEI JIA*, University of Minnesota
SHAOMING XU, University of Minnesota
MICHAEL STEINBACH, University of Minnesota
VIPIN KUMAR, University of Minnesota

There is a growing consensus that solutions to complex science and engineering problems require novel methodologies that are able to integrate traditional physics-based modeling approaches with state-of-the-art machine learning (ML) techniques. This paper provides a structured overview of such techniques. Application areas for which these approaches have been applied are summarized, then classes of methodologies used to construct physics-guided ML models and hybrid physics-ML frameworks are described. We then provide a taxonomy of these existing techniques, which uncovers knowledge gaps and potential crossovers of methods between disciplines that can serve as ideas for future research.

CCS Concepts: • General and reference → Surveys and overviews; • Computing methodologies → Machine learning.

Additional Key Words and Phrases: physics-guided, neural networks, deep learning, physics-informed, theory-guided, hybrid, knowledge integration

ACM Reference Format:
Jared Willard, Xiaowei Jia, Shaoming Xu, Michael Steinbach, and Vipin Kumar. 2020. Integrating Physics-Based Modeling With Machine Learning: A Survey. 1, 1 (July 2020), 34 pages. <https://doi.org/10.1145/1122445.1122456>

1 INTRODUCTION

Machine learning (ML) models, which have already found tremendous success in commercial applications, are beginning to play an important role in advancing scientific discovery in domains traditionally dominated by mechanistic (e.g., first principle) models [22, 112, 117, 126, 127, 138, 214, 261]. The use of ML models is particularly promising in scientific problems involving processes that are not completely understood, or where it is computationally infeasible to run mechanistic models at desired resolutions in space and time. However, the application of even the state-of-the-art black box ML models has often met with limited success in scientific domains due to their large data requirements, inability to produce physically consistent results, and their lack of generalizability to out-of-sample scenarios [126]. Given that neither an ML-only nor a scientific knowledge-only

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*Both authors contributed equally to this research.

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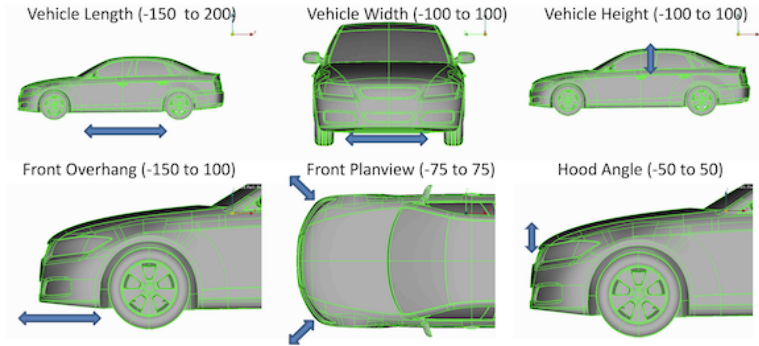
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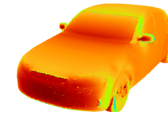
<https://doi.org/10.1145/1122445.1122456>

• Vol. 1, No. 1, Article . Publication date: July 2020.

Designing a car's shape: what does the aerodynamicist need to know?



surface pressure



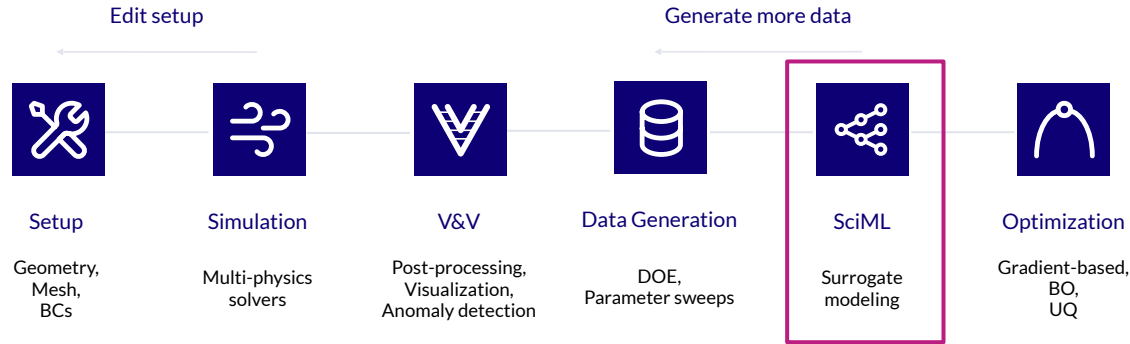
streamwise velocity on the symmetry plane

How much drag? Drag coefficient C_d → range and energy use

Is the lift balanced? Lift C_l and pitching moment C_m → front/rear axle load, high-speed stability

Where do the forces come from? Surface pressure C_p and the flow around the car → where the flow separates, and where to change the shape

One CFD run takes hours, and a design loop needs 1000s of them.



Two ways to get speed, and gradients:

	Neural surrogate	Differentiable solver
Idea	Learn the map design → fields from simulation data	Make the solver itself a program that gradients flow through
Cost	Expensive offline (data + training), ms at inference	Cost of a solve, plus a backward pass
Trust	Approximate: errors from data coverage, architecture and training; no built-in error estimate	Physics exact up to discretization

Today

This talk!

Lightning talk 1 (Niels), tutorial

Where does the physics go?

The PINN loss as the example:

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N_d} \sum_i \|u_\theta(x_i) - u_i\|^2}_{\text{data}} + \lambda_r \underbrace{\frac{1}{N_r} \sum_j \|\mathcal{N}[u_\theta](x_j) - f(x_j)\|^2}_{\text{PDE residual}} + \lambda_b \underbrace{\frac{1}{N_b} \sum_k \|\mathcal{B}[u_\theta](x_k) - g(x_k)\|^2}_{\text{boundary/initial conditions}}$$

	Constrained	Penalized/informed	Architecture
How	Forced to satisfy the physical model	Physics as a penalty in the loss	Physical properties built into the model
Example	Impose known solutions of the governing equation (e.g. hard BCs)	Add the governing equation; small deviations allowed (PINNs)	Encode general properties: symmetry, locality, conservation, multiscale
Strictness	Exact	Soft	Structural

- PINNs in practice: Lightning talk 2 (Jakob)
- The rest of this talk: the architecture column

From learning a solution to learning the solver

Learn the operator $\mathcal{G} : u \mapsto \mathcal{G}(u)$ (coefficients, BCs, geometry $\rightarrow$ solution field), independently of the discretization.

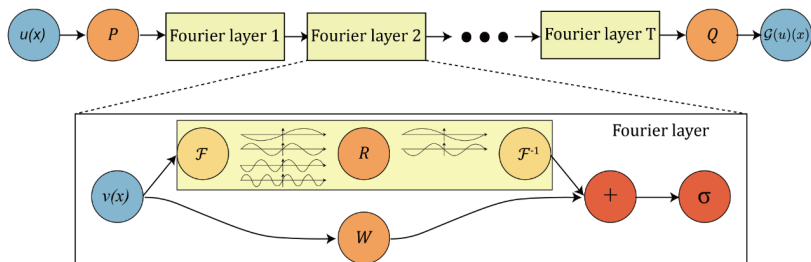
Fourier Neural Operator (FNO)

$$v_{\ell+1}(x) = \sigma(Wv_{\ell}(x) + \mathcal{F}^{-1}(R_{\phi} \cdot \mathcal{F}v_{\ell})(x))$$

Global convolution in Fourier space, truncated to low modes

Fast on regular grids

Needs a grid; struggles with complex geometry



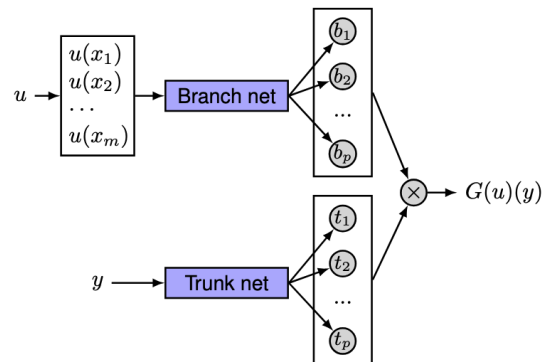
DeepONet

$$\mathcal{G}(u)(y) \approx \sum_{k=1}^p b_k(u) t_k(y)$$

Global convolution in Fourier space, truncated to low modes

Arbitrary query points

Basis is global, not adapted to the geometry



What industrial CAE actually looks like

Steady (BVP): the boundary shapes the whole domain → global reach in one forward pass

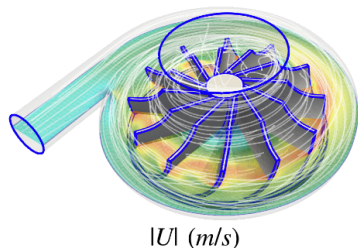
Evolution (IVP/IBVP): march from an initial state → stable long rollouts

Multiphysics: coupled fields across scales and solvers → information across interfaces

→ **Unstructured meshes** of 10^5 – 10^8 cells, and **hundreds of samples** (not millions)

Centrifugal pump

- 300 samples
- ~269MB, 1.4M cells, ~80 GB in total



Car aero dynamics (DrivAerML dataset)

- 500 samples
- ~0.6 GB, ~9M points/~45 GB, ~160M cells, ~31 TB in total



Ashton et al. ([arXiv:2408.11969](https://arxiv.org/abs/2408.11969))

Graph networks see the neighbours, transformers see the big picture

Local: MeshGraphNet (MGN)

Mesh = graph; message passing along edges

Physics in the architecture: locality, permutation invariance, relative coordinates (a learned stencil)

✓ Sharp local detail

✗ Information moves one hop per layer

Global: latent-token transformers

Compress N mesh points into $L \ll N$ latent tokens, process them, write back

Transolver, Perceiver/UPT, Surya

✓ Global reach in one layer

✗ Behaves like a low-pass filter: smooths fine scales

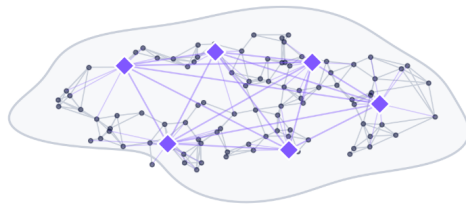
message passing

each node updates from its mesh neighbours



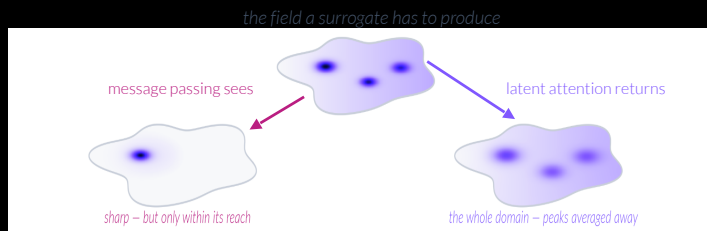
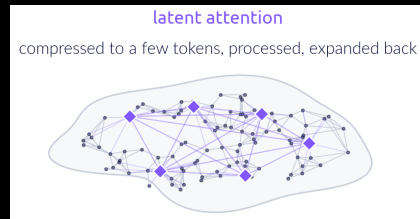
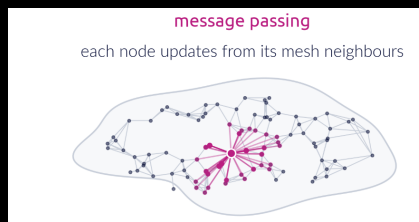
latent attention

compressed to a few tokens, processed, expanded back



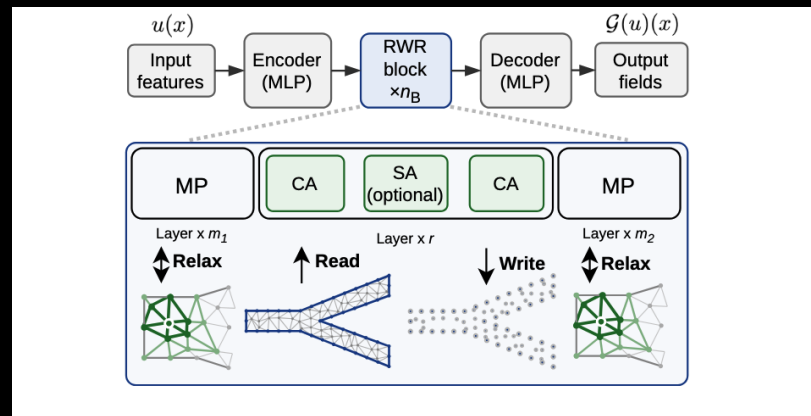
Read-Write-Relax (RWR), a neural PDE surrogate with local AND global processing

RWR interleaves local and global processors similar to a multigrid cycle: one correcting low-frequency error, the other high-frequency error.



sharp, but only within its reach

the whole domain, but peaks averaged away



Can a surrogate stand in for CFD in a car-aero design loop?

The Read-Write-Relax (RWR) surrogate was trained on the DrivAerML dataset containing ~500 morphed variants of the DrivAer car (Ashton et al. 2024)

Question

Why it matters

Fields

Are the surface fields right everywhere?

Surface pressure produces most of the force

Profiles

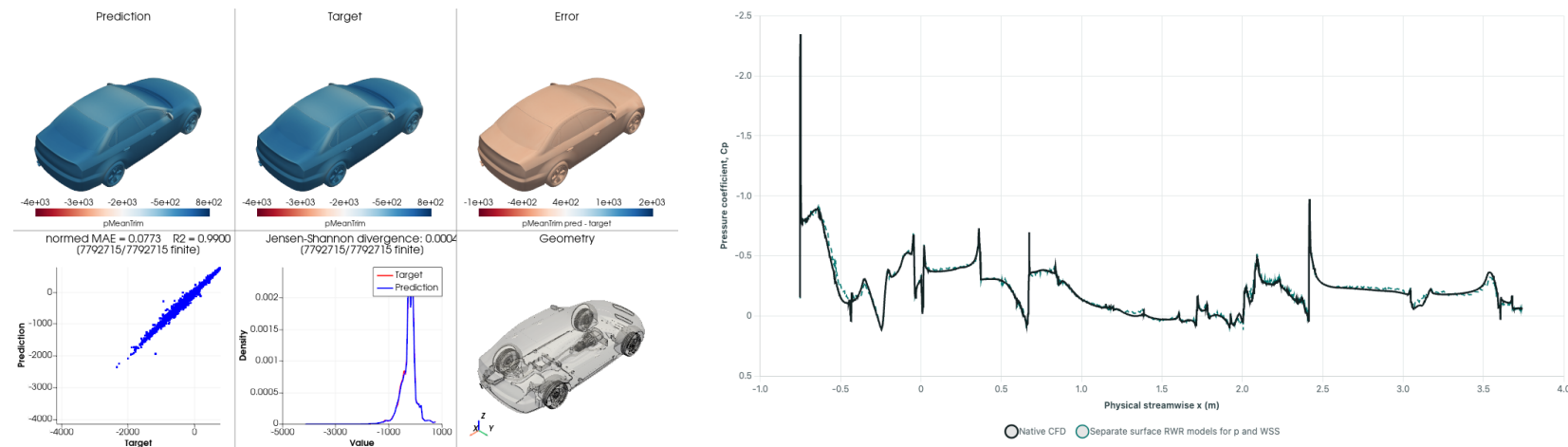
Is C_p right along body sections?

Comparable with wind-tunnel probes

Forces

Do C_d , C_l , C_m follow design changes car to car?

Drag $\rightarrow$ range; lift balance $\rightarrow$ handling



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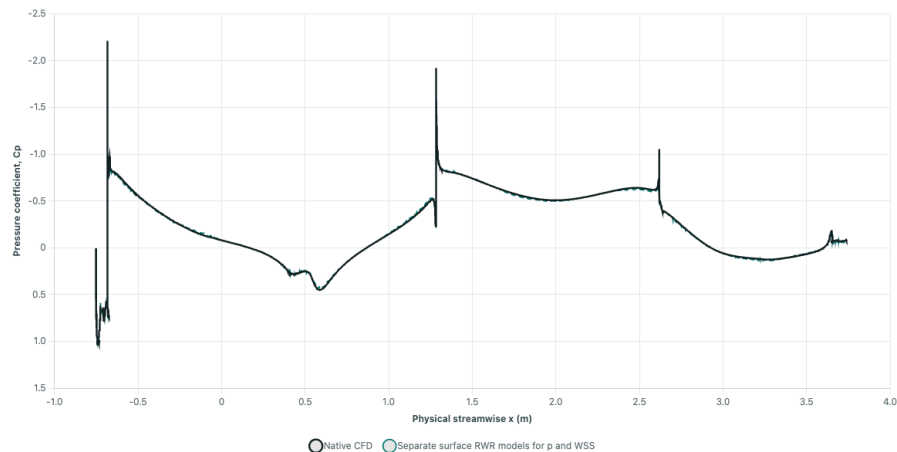
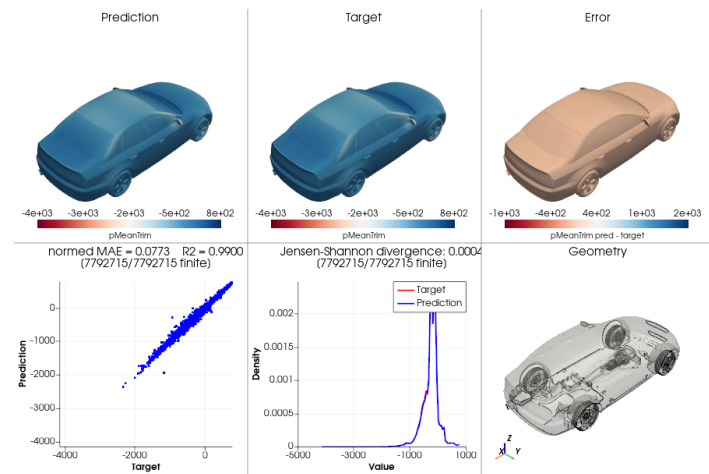
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Takeaways

1. Physics can go into the loss, the constraints, or the architecture (the architecture route scales best)
2. Neural operators learn solvers, not solutions
3. Both global reach and local resolution are needed on real-world meshes
4. Validate on what engineers care about: forces, profiles, not bulk error alone.

Many successful SciML methods are inspired by what made solvers great (RWR = multigrid-inspired)

Next in this session:

⚡ Differentiable electromagnetics, learning inside the solver (Niels)

⚡ Sensing with PINNs, the loss route (Jakob)